Use Pythagoras’ theorem to solve problems involving the side lengths of right-angled triangles

Learning intention

- use Pythagoras’ theorem to solve problems involving the side lengths of right-angled triangles.

Teach and model

- identifying Pythagorean triples, such as (3,4,5), (5,12,13), (7, 24, 25) and (8,15, 17).
- investigating how Pythagoras' theorem can be applied to determine the distance between two points in the plane.
- recognising the relationship between the squares of lengths of sides for different types of triangles: right-angled, acute or obtuse.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.